THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A square and an equilateral triangle have equal perimeters. If the square's side is 15 cm, the triangle's side (cm) is: 1

(a) 25 **(b)** 35 **(c)** 20 **(d)** 15

2. A rectangular field 30 m by 16 m is divided by a diagonal into two triangles. The area of one triangle (m^2) is: 1

(a) 270 **(b)** 210 **(c)** 240 **(d)** 256

3. A picture 40 cm by 25 cm is mounted on a frame of width 4 cm all around. The area of the frame alone (cm^2) is: 1

(a) 584 **(b)** 580 **(c)** 600 **(d)** 588

4. A wire in the shape of a square of side 25 cm is straightened and bent into a rectangle of length 32 cm. The breadth (cm) is: 1

(a) 32 **(b)** 18 **(c)** 11 **(d)** 7

5. Tiling a rectangular floor of length 12 m at ₹80 per m^2 costs ₹6720. Its breadth (m) is: 1

(a) 19 **(b)** 7 **(c)** 84 **(d)** 14

6. Out of all rectangles with perimeter 40 cm having whole-number sides, the greatest area (cm^2) is: 1

(a) 90 **(b)** 110 **(c)** 100 **(d)** 120

7. The sides of a triangle are 17 cm, 25 cm and 26 cm. Its perimeter (cm) is: 1

(a) 68 **(b)** 85 **(c)** 67 **(d)** 69

8. A square park of side 30 m has a footpath 1 m wide all around outside it. The area of the footpath (m^2) is: 1

(a) 132 **(b)** 120 **(c)** 124 **(d)** 128

9. **Assertion (A):** A rectangle has area 48 cm^2 and length 8 cm. Its breadth is 6 cm. 1

Reason (R): Breadth = area x length.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A rectangle and a square with the same perimeter always have the same area. 1
Reason (R): The area of a square of side a is a x a.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A rectangular field is 50 m by 20 m. Find the cost of levelling the field at ₹4 per m^2 and fencing at ₹15 per metre. 2

12. A carpet 6 m by 5 m is laid in a room 9 m by 7 m. Find the area of the uncovered floor. 2

13. The length and breadth of a rectangle are in the ratio 4:3 and its perimeter is 98 cm. Find the sides. 2

14. A rectangular field is 60 m by 45 m. Find the cost of levelling the field at ₹3 per m^2 and fencing at ₹10 per metre. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangular plot 60 m by 35 m is to be fenced with 3 rounds of barbed wire costing ₹10 per metre. Find the total cost. Also find the cost of ploughing it at ₹3 per m^2. 3

16. A square of side 20 cm and a rectangle of length 24 cm and breadth 16 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

17. A floor 9 m by 6 m is to be covered with tiles each 30 cm by 15 cm. Find the number of tiles. 3

OR

A rectangular plot 45 m by 20 m is to be fenced with 4 rounds of barbed wire costing ₹8 per metre. Find the total cost. Also find the cost of ploughing it at ₹6 per m^2. 3

18. A square of side 18 cm and a rectangle of length 23 cm and breadth 13 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

OR

A floor 10 m by 4 m is to be covered with tiles each 50 cm by 20 cm. Find the number of tiles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A school courtyard is 12 m long and 9 m wide. It is paved with square slabs of side 3 m. 5
(i) Find the area of the courtyard.
(ii) Find the area of one slab.
(iii) How many slabs are needed?
(iv) Each slab costs ₹450. Find the total cost and the cost per square metre.

20. A rectangle with length 16 cm and breadth 9 cm has a triangular piece with base 16 cm and height 6 cm cut from one side. Find the area of the remaining piece and the fraction of the whole that is left. 5

OR

A rectangular park 90 m by 60 m has two paths each 5 m wide crossing in the middle, one parallel to each side. Find the area of the paths and the area of the rest of the park. 5

END OF QUESTION PAPER — Total: 40 Marks